Pre-Analysis Plan

Instructions: First, we define the main independent variable for H1-H4. Second, we define dependent variables (DVs), exclusion criteria, specification of regression model, and interpretation of results for each individual hypothesis. Finally, we include a table of control variables for all hypotheses at the of the document.

A. Construction of the Independent Variable

IV1: Construction of the "Has a Daughter" Variable

To identify whether a respondent has at least one daughter, we use data from the SOEP birth biography module (biobirth) on pages 36–38. Respondents are asked to report key details for up to ten biological children, including:

- Number of children (biokids– Number Of Births From Biobirth: Do you have or had children? If so, how much? // Yes ... children)
- The gender of each child (1 = male, 2 = female). The gender of each child is recorded in variables: kidsex1, kidsex2, ..., kidsex10.

The construction will be conditioned on the variable biokids, which indicates the total number of children a respondent reports. biokids is located in the same module on page 36 of biobirth dataset.

The binary variable HasDaughter is defined as follows:

- 1 if the respondent has at least one daughter (kidsexi == 2)
- 0 if the respondent has only sons
- Missing (.) if the respondent reports having no children (biokids == 0)

B. Information for each individual hypothesis

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality:

(X for respondents with daughters) > (X for respondents without daughters).

Variables

Which variables will you include? How will these variables be coded – will you use them the way they are or recode them, and if so, how?

Dependent Variables:

Original question: I will read you a series of statements. To what degree do you personally agree with each statement?

V1: Children below the age of 6 suffer if their mother works (1- Do not agree at all to 7- Completely agree). Reverse code for simplicity

Variable name: bip_197_04

V2: A child under the age of three suffers if his or her mother works (1- Do not agree at all to 7- Completely agree). Reverse code for simplicity

Variable name: bip_197_05

V3: It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally (1- Do not agree at all to 7- Completely agree).

Variable name: bip_197_06

DV_H1: Average of V1, V2 and V3.

Note: these three gender attitude questions only available for 2018 individual questionnaire (pl dataset, Question 197, page 66). The 2018 questionnaire: https://paneldata.org/soep-core/instruments/soep-core-2018-pe/197

Inclusion/exclusion criteria

Are there any observations you will exclude? If yes, which? Which years will you include/what time period will you study?

Exclusion criteria: must be double parent family (pld0133="Yes"), no childfree people included. The variable biokids in biobirth dataset indicate the total number of children a respondent reports. biokids > 0 is used as a condition to restrict the sample, which means we only have a sample people who has at least one child. Gender attitude questions were only included in 2018. Hence, we are including only observations from 2018 for this hypothesis.

Exact analysis

Please write down the exact specification of the regression model, including the names of the dependent and independent variables and whether you use robust standard errors or something else.

$DV_H1_i = \beta_o + \beta_1 \cdot HasDaughter_i + + \gamma' Z_i + \varepsilon_i$

Where X is either DV_H1 as defined above. And Z denotes a vector of control variables (described below).

Interpretation of results

Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

Statistical Test

Estimate β_1 using censored TOBIT regressions since the DV_H1 is censored between 0 and 7. Use t-test to assess significance as reported by regression estimation output.

Interpretation

If $\beta_1 > 0$ and statistically significant, this supports H1.

H2. Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences: (Y for respondents with daughters) > (Y for respondents without daughters).

Variables

Which variables will you include? How will these variables be coded – will you use them the way they are or recode them, and if so, how?

Dependent Variables:

DV_H2: Left-Right political preferences. Coded from 0 to 10 with zero being extremely far left and 10 being extremely far right.

- ➔ In politics people often talk about “left” and “right” when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?
- ➔ Variable name: bjp_171

Extracted from: question 171, soep 2019 questionnaire: https://paneldata.org/soep-core/instruments/soep-core-2019-pe-lgb/171#

Inclusion/exclusion criteria

Are there any observations you will exclude? If yes, which? Which years will you include/what time period will you study?

Exclusion criteria: must be double parent family (pld0133=”Yes”), no childfree people included. The variable biokids in biobirth dataset indicate the total number of children a respondent reports. biokids > 0 is used as a condition to restrict the sample, which means we only have a sample people who has at least one child. We will include only observations from 2019.

Exact analysis

Please write down the exact specification of the regression model, including the names of the dependent and independent variables and whether you use robust standard errors or something else.

$DV_H2_i = \beta_0 + \beta_1 \cdot HasDaughter_i + \gamma' Z_i + \varepsilon_i$

Where DV_H2 is bjp_171 as defined above. And Z denotes a vector of control variables (described below).

Interpretation of results

Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

Statistical Test

Estimate β_1 using a Tobit regression since DV_H2 is censored between 0 and 10.

Interpretation

If $\beta_1 < 0$ and statistically significant, this supports H2 (since higher bjp_171 means more right-leaning).

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household:

(Z for respondents with daughters) > (Z for respondents without daughters).

Variables

Which variables will you include? How will these variables be coded – will you use them the way they are or recode them, and if so, how?

Dependent Variables:

V1: Male share of hours spent on childcare.

V2: Male share of hours spent on housework.

DV_H3: Inequality index for hours spent on childcare/housework.

Step by step to compute DV_H3:

1. Extract number of hours spent on childcare/housework from individual questionnaire 2018 (question 17).
2. Partners can be matched using plk0001 (never changing person ID) where the respondent reports the name of the partner and this connected to the partner's person ID (question 194 from individual questionnaire in 2018). We include only

respondents who report having a partner that lives in the same household using variable pld0133 (question 193 from individual questionnaire in 2018).

3. Merge partner information and create wide format dataset (one couple = 1 row) that includes respondent's id, respondent's sex, respondent's hours spent on childcare/housework, partner's id, partner's hours spent on childcare/housework and other relevant control variables. Now, for each respondent all this information should be in the same row for each couple.
4. Generate $total_childcare = childcare_f + childcare_m$ and $total_housework = housework_f + housework_m$ and $male_{share} = \frac{housework_m + childcare_m}{(total_housework + total_childcare)}$
5. For DV_H3, we calculate DV_H3= Inequality_index_ = absolute (0.50 – male_share).

Inclusion/exclusion criteria

Are there any observations you will exclude? If yes, which? Which years will you include/what time period will you study?

Exclusion criteria must be double parent family (pld0133="Yes"), no childfree people included. The variable biokids in biobirth dataset indicate the total number of children a respondent reports. biokids > 0 is used as a condition to restrict the sample, which means we only have a sample people who has at least one child. We will only include observations for 2018.

Exact analysis

Please write down the exact specification of the regression model, including the names of the dependent and independent variables and whether you use robust standard errors or something else.

$DV_H3_h = \beta_0 + \beta_1 \cdot HasDaughter_h + Z_h' \cdot \gamma + \varepsilon_h$

Where, DV_H3 is the inequality index in terms of childcare and housework in the household h. Z_h is a vector of household-level control variables (described below).

Interpretation of results

Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

Statistical Test

Estimate β_1 in a Tobit Regression (since inequality index is censored between 0 - 0.5).

Interpretation

If $\beta_1 < 0$ and significant, this supports H3.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality

measured by attitudes). That is, denoting X as a measure of support for gender equality:

(X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters)

Variables

Which variables will you include? How will these variables be coded – will you use them the way they are or recode them, and if so, how?

Dependent Variable: This will be the same as in H1

V1: Children below the age of 6 suffer if their mother works (1- Do not agree at all to 7- Completely agree). Reverse code for simplicity.

Variable name: bip_197_04

V2: A child under the age of three suffers if his or her mother works (1- Do not agree at all to 7- Completely agree). Reverse code for simplicity

Variable name: bip_197_05

V3: It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally (1- Do not agree at all to 7- Completely agree).

Variable name: bip_197_06

DV_H4=DV_H1: Average of V1, V2 and V3 (reversed).

Original question: I will read you a series of statements. To what degree do you personally agree with each statement?

Note: these three gender attitude questions only available for 2018 individual questionnaire (pl dataset, Question 197, page 66). The 2018 questionnaire: https://paneldata.org/soep-core/instruments/soep-core-2018-pe/197

Inclusion/exclusion criteria

Are there any observations you will exclude? If yes, which? Which years will you include/what time period will you study?

Exclusion criteria must be double parent family (pld0133="Yes"), no childfree people included. The variable biokids in biobirth dataset indicate the total number of children a respondent reports. biokids > 0 is used as a condition to restrict the sample, which means we only have a sample people who has at least one child. Gender attitude questions were

only included in 2018. Hence, we are including only observations from 2018 for this hypothesis.

Exact analysis

Please write down the exact specification of the regression model, including the names of the dependent and independent variables and whether you use robust standard errors or something else.

$DV_H4_i = \beta_0 + \beta_1 \cdot HasDaughter_i + \beta_2 \cdot SexMale_i + \beta_3 \cdot (HasDaughter_i \times SexMale_i) + \gamma' Z_i + \varepsilon_i$

Where DV_H4_i is the DV_H4 = DV_H1 as described above. SexMale is the Sex of the responder – either 1 male (father) or 0 female (mother)

Interpretation of results

Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

Statistical Test

Key coefficient is β_3 (interaction effect). Estimate using tobit since the values of the DV_H4 is censored between 1-7.

Interpretation

If $\beta_3 > 0$ and significant, this supports H4 (having a daughter has a greater effect for men than women).

C. List of Control Variables

Control Variables for H1, H2 and H4. Regressions at the individual level.

Control Variable	Justification	Dataset (Source)	Variable Name	Page Number (Approx.)
SexMale	Key moderator for heterogeneous effects (especially for H4)	biobirth	Sex (recoded as 0 if Female and 1 if Male)	10
Age	Age influences attitudes and political preferences	pgen	Use gebjahr (birth year) to compute age. Survey year minus gebjahr.	11

Control Variable	Justification	Dataset (Source)	Variable Name	Page Number (Approx.)
Education Level	Education correlates with gender attitudes and political leaning	pgen	pgisced11 – ISCED-2011-Classification	53
Income	Income may shape both attitudes and gender roles	pgen	pglabnet	12
Religiosity	Religiosity impacts views on gender roles and politics	pl	bjp/bjp_194_01 (variable from 2019; name of variable changes every year), recode to 1 if respondent belongs to any religious community, 0 otherwise.	34
Number of Children	Ensures treatment group is actually parents	biobirth	biokids, kidsex1–kidsex10	36–38
Age of youngest child	Youngest child requires more care	biobirth	kidgeb01-kigeb10 for year of birth of child	24
Marital Status	Marital status affects household dynamics and attitudes	pgen	Pgpartz, recoded 1 if respondent has partner or spouse, 0 otherwise	9
Partner's Employment	Partner’s employment influences division of labor at home	pgen	pgemplst via pgpartnr	19
Partner's Education	Partner’s education may impact household gender norms	pgen	pgisced11 via pgpartnr	9 and 53

Control Variables for H3. Regressions at the household (dyadic) level.

Control Variable	Why It Matters	SOEP Dataset	Construction Notes
Number of Children	More children may increase total care needs and affect division of labor.	biobirth	Use biokids variable.
Age of Youngest Child	Younger children increase care intensity; may influence role specialization.	biobirth	Use childbirth years (kidgeb01-10); subtract from survey year; take minimum.
Total Household Income	Higher income may allow outsourcing or reduce traditional constraints.	pgen	Sum personal net incomes of both partners using pglabnet.
Education of Male/Female Partner	Higher education is correlated with egalitarian norms.	pgen	Use ISCED or years of schooling of both; take max or create dummies using the variable pgisced11
Both Partners Employed (Binary)	Employment status affects time available for housework and earnings.	pgen	Use pgemplst and generate binary indicator if both partners are employed.
Age of Male/Female Partner	Age may proxy for generational attitudes and physical ability.	biobrth	Use gebjahr (birth year) to compute age.